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Method for displaying stereoscopic scene and electronic device

Abstract

A method for displaying a stereoscopic scene and an electronic device are provided. The electronic device includes a displaying unit, an inputting unit, a stereo scene generating unit, a controlling unit and an imaging unit. The inputting unit includes a hot key. If the hot key is pressed, the stereoscopic scene generating unit generates the stereoscopic scene comprising a plurality of frames. The controlling unit is used for receiving a human interface device (HID) signal from the inputting unit, and converting the HID signal into an angle signal. The imaging unit is used for displaying one of the frames of the stereoscopic scene on the displaying unit according to the angle signal.

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Background/Summary

(1) This application claims the benefit of Taiwan application Serial No. 112123379, filed Jun. 21, 2023, the subject matter of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The invention relates in general to a displaying method and an electronic device, and more particularly to a method for displaying a stereoscopic scene and an electronic device.

Description of the Related Art

(3) Along with the rapid development in the multimedia and gaming industry, the user can operate stereoscopic scene with a specifically matched remote device. For instance, the user can view different angles of virtual frames through the use of a head mounted display. However, without the specifically matched remote device, the user will be unable to operate the stereoscopic scenes.

SUMMARY OF THE INVENTION

(4) The invention is directed to a method for displaying a stereoscopic scene and an electronic device. Through the use of HID angle conversion technology, the HID signal of the inputting unit can be converted into an angle signal, so that the stereoscopic scene can be operated with various inputting units. Besides, through the use of a computational balancing mechanism, the AI inference procedure can be oriented to the artificial intelligence acceleration processor, so that the loading of the CPU or the GPU can be reduced.

(5) According to a first aspect of the present invention, a method for displaying a stereoscopic scene is provided. The displaying method includes the following steps. Whether a hot key of an inputting unit is pressed is determined: if the hot key is pressed, the stereoscopic scene comprising a plurality of frames is generated. A human interface device (HID) signal is received from the inputting unit. The HID signal is converted into an angle signal. One of the frames of the stereoscopic scene is displayed on the displaying unit according to the angle signal.

(6) According to a second aspect of the present invention, an electronic device is provided. The electronic device includes a displaying unit, an inputting unit, a stereo scene generating unit, a controlling unit and an imaging unit. The inputting unit includes a hot key. If the hot key is pressed, the stereoscopic scene generating unit generates a stereoscopic scene comprising a plurality of frames. The controlling unit is used for receiving a human interface device (HID) signal from the inputting unit and converting the HID signal into an angle signal. The imaging unit is used for displaying one of the frames of the stereoscopic scene on the displaying unit according to the angle signal.

(7) The above and other aspects of the invention will become better understood with regard to the following detailed description of the preferred but non-limiting embodiment(s). The following description is made with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is an example of operating a particular stereoscopic scene with an electronic device.

(2) FIG. 1B is an example of operating a particular stereoscopic scene with an electronic device.

(3) FIG. 2 is a block diagram of an electronic device according to an embodiment.

(4) FIG. 3 is a flowchart of a method for displaying a stereoscopic scene according to an embodiment.

(5) FIG. 4 is a detailed block diagram of a controlling unit according to an embodiment.

(6) FIG. 5 is a detailed flowchart of step S160 according to an embodiment.

(7) FIG. 6 is a detailed block diagram of a stereoscopic scene generating unit according to an embodiment.

(8) FIG. 7 is a detailed flowchart of step S140 according to an embodiment.

(9) FIG. 8 is a flowchart of a method for displaying a stereoscopic scene according to another embodiment.

(10) FIG. 9 is a detailed flowchart of step S190 according to an embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(11) Referring to FIG. 1A, an example of operating a particular stereoscopic scene SN1 with an electronic device 100 is shown. The stereoscopic scene SN1 can be realized by a space surround image. The stereoscopic scene SN1 can be rotated either manually or automatically to display different directions of an image. When the stereoscopic scene SN1 rotates, the object Oi moves accordingly. As indicated in FIG. 1A, the stereoscopic scene SN1 can be realized by the scene of a racing game, and by default setting, the stereoscopic scene SN1 is operated with a game steering wheel CT1. When the game steering wheel CT1 rotates, an angle operating signal A1 can be inputted to rotate the stereoscopic scene SN1. Ordinary peripheral inputting units (such as game joystick CT2) only allow the input of human interface device (HID) signal H2, which does not match the racing game, therefore cannot operate the stereoscopic scene SN1.

(12) In the present embodiment, through the use of HID angle conversion technology, the HID signal H2 of whatever peripheral inputting units (such as game joystick CT2) can be converted into an angle signal for operating the stereoscopic scene SN1.

(13) Referring to FIG. 1B, an example of operating a particular stereoscopic scene SN2 with an electronic device 100 is shown. The stereoscopic scene SN2 can be realized by a three-dimensional figure. The stereoscopic scene SN2 can be rotated either manually or automatically to display directions of an image. In the present embodiment, through the use of HID angle conversion technology, the HID signal H3 of whatever peripheral inputting units (such as game joystick CT3) can be converted into an angle signal for operating the stereoscopic scene SN2.

(14) Besides, during the process of displaying and operating the stereoscopic scenes SN1 and SN2, the central processing unit (CPU) 111 or the graphics processing unit (GPU) 112 experiences a heavy loading. Meanwhile, if the object Oj additionally requires recognition inference, the CPU 111 or the GPU 112 will be overloaded. In the present embodiment, through the use of a computational balancing mechanism, the AI inference procedure is oriented to the artificial intelligence acceleration processor 113, so that the loading of the CPU 111 or the GPU 112 can be reduced.

(15) Referring to FIG. 2, a block diagram of an electronic device 100 according to an embodiment is shown. The electronic device 100 includes the said CPU 111, the said GPU 112, the said artificial intelligence acceleration processor 113, a displaying unit 120, an inputting unit 130, a stereoscopic scene generating unit 140, a plurality of registers 150k, a controlling unit 160, an imaging unit 170 and an inference procedure translation unit 190. The artificial intelligence acceleration processor 113 can be realized by a neural-network processing unit (NPU) or an intelligence processing unit (IPU). The artificial intelligence acceleration processor 113 is different from the GPU 112. The artificial intelligence acceleration processor 113 is equipped with an in memory computing function or an in memory searching structure and is applicable to the AI inference procedure.

(16) The displaying unit 120 can be realized by a liquid crystal display panel, an OLED panel, an e-paper, a head mounted display, an AR glasses or a projector. The inputting unit 130 can be realized by a keyboard, a mouse or a joystick. The inputting unit 130 is not limited to a single device; the inputting unit 130 can also be formed of a plurality of devices. The stereoscopic scene generating unit 140, the controlling unit 160, the imaging unit 170, and the inference procedure translation unit 190 are used for executing various procedures, such as processing procedure, computing procedure, analytic procedure, and judgment procedure, and can be realized by a circuit, a circuit board, a chip, or a storage device for storing program code. The register 180k is used for temporarily storing data, and can be realized by a memory or a storage block of a hard disc. In the present embodiment, through the use of HID angle conversion technology, the controlling unit 160 can convert an HID signal Hn of the inputting unit 130 into an angle signal An for operating the

stereoscopic scene SNm. Also, in the present embodiment, through the use of a computational balancing mechanism, the inference procedure translation unit **190** can orient the AI inference procedure to the artificial intelligence acceleration processor **113**, so that the loading of the CPU **111** or the GPU **112** can be reduced. The operations of each element are disclosed below with accompanying drawings.

(17) Referring to FIG. 3, a flowchart of a method for displaying a stereoscopic scene SNm according to an embodiment is shown. In step **S110**, a hot key **131** of the inputting unit **130** is pressed is determined by the controlling unit **160**. The hot key **131** is a pre-defined key or a pre-defined combination of a plurality of keys. When the user requires the stereoscopic scene SNm, the user can press the hot key **131**. In an embodiment, the hot key **131** can be realized by a particular function key disposed on the keyboard. If the hot key **131** is pressed, the method proceeds to step **S140**.

(18) In step **S140**, the stereoscopic scene SNm is generated by the stereoscopic scene generating unit **140**. The stereoscopic scene SNm consists of a plurality of frames FMt. The stereoscopic scene SNm can be realized by a particular space surround image (such as the stereoscopic scene SN1 of FIG. 1); the stereoscopic scene SNm can be a particular three-dimensional figure or object (such as the stereoscopic scene SN2 of FIG. 2).

(19) Then, the method proceeds to step **S150**, an HID signal Hn is received from the inputting unit **130** by the controlling unit **160**. The device which transmits the HID signal Hn can be different from the device equipped with the hot key **131**. For instance, the device which transmits the HID signal Hn can be an external joystick, a mouse, an external touch panel, a game handle or a similar device; the device equipped with the hot key **131** can be realized by the keyboard or in-built touch panel of the device.

(20) Then, the method proceeds to step **S160**, the HID signal Hn is converted into the angle signal An by the controlling unit **160**.

(21) Refer to FIG. 4 and FIG. 5. FIG. 4 is a detailed block diagram of a controlling unit according to an embodiment. FIG. 5 is a detailed flowchart of step **S160** according to an embodiment. The controlling unit **160** includes an HID value retrieving component **161**, a normalization component **162** and an angle conversion component **163**. Step **S160** includes steps **S161** to **S163**. In step **S161**, an HID value Vn of the HID signal Hn is obtained by the HID value retrieving component **161** of the controlling unit **160**. For instance, when the HID signal Hn is received from a press key, the HID value Vn can be 0 or 1; when the HID signal Hn is received from a joystick mushroom head, the HID value Vn can be decimal values in the X direction and the Y direction; when the HID signal Hn is received from the touch panel, the HID value Vn can be the coordinate value of the X axis and Y axis.

(22) Then, the method proceeds to step **S162**, the HID value Vn is normalized by the normalization component **162** of the controlling unit **160** to obtain a normalized value Vn'. For instance, the normalization component **162** normalizes the HID value Vn as a normalized value Vn' between -1 to 1. For instance, the maximum possible value of HID value Vn is normalized as 1; the minimum possible value of HID value Vn is normalized as -1. Or, the normalization component **162** can be normalized as two normalized values Vn' in the X direction and the Y direction according to the X axis and the Y axis, respectively.

(23) Then, the method proceeds to step **S163**, the normalized value Vn' is multiplied by 90° by the angle conversion component **163** of the controlling unit **160** to obtain the angle signal An. That is, the normalized value Vn' being "1" will be converted to the angle signal An whose value is "90°"; the normalized value Vn' whose value is "-1", will be converted to the angle signal An whose value is "-90°"; the normalized value Vn' whose value is "0.5" will be converted to the angle signal An whose value is "45°", and the rest can be obtained by the same analogy.

(24) Through steps **S161** to **S163**, the controlling unit **160** can convert the HID signal Hn into the angle signal An.

(25) Next, the method proceeds to step **S170** of FIG. 3, a particular frame **FMt'** of the stereoscopic scene **SNm** is displayed on the displaying unit **120** by the imaging unit **170** according to the angle signal **An**. During the operation process of the user, different **HID** signals **Hn** generated by the inputting unit **130** can correspond to different angle signals **An**; the frame **FMt'** of the stereoscopic scene **SNm** that corresponds to a particular angle can be displayed on the displaying unit **120**. Thus, through steps **S110** to **S170**, the user can display the stereoscopic scene **SNm** with any inputting unit **130**.

(26) Refer to FIG. 6 and FIG. 7. FIG. 6 is a detailed block diagram of a stereoscopic scene generating unit according to an embodiment. FIG. 7 is a detailed flowchart of step **S140** according to an embodiment. The stereoscopic scene generating unit **140** includes a searching component **141** and a scene register **142**. In an embodiment, when the imaging unit **170** performs imaging, graphics information can be received from an external server connected to an application unit **AP**. The electronic device **100** is without the stereoscopic scene **SNm**, therefore requires the stereoscopic scene generating unit **140** for generating the stereoscopic scene **SNm**.

(27) In step **S141**, the searching component **141** of the stereoscopic scene generating unit **140** searches the frame **FMt** necessary for the stereoscopic scene **SNm** from the frame register **150k**. During the operation of the application unit **AP**, the frames need to be decoded and rendered, and different angles of the frame **FMt** are temporarily stored in the frame register **150k**, so that the searching component **141** can collect different angles of frame **FMt** from the registers **150k**.

(28) Then, the method proceeds to step **S142**, the frames **FMt** are temporarily stored in the scene register **142** of the stereoscopic scene generating unit **140** to form the stereoscopic scene **SNm**.

(29) Through steps **S141** to **S142**, the stereoscopic scene generating unit **140** can generate the stereoscopic scene **SNm**.

(30) Referring to FIG. 8, a flowchart of a method for displaying a stereoscopic scene **SNm** according to another embodiment is shown. The displaying method illustrated in FIG. 8 further includes steps **S180** to **S190**. In steps **S110** to **S170**, the process of displaying and operating the stereoscopic scene **SNm** needs to be computed with the CPU **111** or the GPU **112**. Meanwhile, the execution of the AI inference procedure will make the CPU **111** or the GPU **112** overloaded. In the present embodiment, through the use of a computational balancing mechanism of steps **S180** to **S190**, the AI inference procedure is oriented to the artificial intelligence acceleration processor **113**, so that the loading of the CPU **111** or the GPU **112** can be reduced.

(31) In step **S180**, whether an inference procedure needs to be performed is determined by the inference procedure translation unit **190**. If the inference procedure needs to be performed, the method proceeds to step **S190**.

(32) In step **S190**, the computational balancing mechanism is activated by the inference procedure translation unit **190** to control the electronic device **100** to perform the inference procedure by the artificial intelligence acceleration processor **113**.

(33) Referring to FIG. 9, a detailed flowchart of step **S190** according to an embodiment is shown. step **S190** includes steps **S191** to **S193**. In step **S191**, an inference model **MD** is loaded to a computing register **191** by the inference procedure translation unit **190**.

(34) Next, In the method proceeds to **S192**, the address of the computing register **191** is directed to the artificial intelligence acceleration processor **113** by the inference procedure translation unit **190**.

(35) Then, the method proceeds to step **S193**, the inference model **MD** is executed by the artificial intelligence acceleration processor **193** to perform the inference procedure. Through steps **S191** to **S193**, during the process of displaying and operating the stereoscopic scene **SNm**, the AI inference procedure will automatically be oriented to the artificial intelligence acceleration processor **193**, so that the loading of the CPU **111** or the GPU **112** will not be increased.

(36) According to the above embodiments, through the use of **HID** angle conversion technology, the **HID** signal **Hn** of the inputting unit **130** can be converted into an angle signal **An** for operating the stereoscopic scene **SNm**. Besides, according to the above embodiments, through the use of a

computational balancing mechanism, the AI inference procedure can be oriented to the artificial intelligence acceleration processor **113**, so that the loading of the CPU **111** or the GPU **112** can be reduced.

(37) While the invention has been described by way of example and in terms of the preferred embodiment(s), it is to be understood that the invention is not limited thereto. On the contrary, it is intended to cover various modifications and similar arrangements and procedures, and the scope of the appended claims therefore should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements and procedures.

Claims

1. A method for displaying a stereoscopic scene, comprising: determining whether a hot key of an inputting unit is pressed; if the hot key is pressed, generating the stereoscopic scene comprising a plurality of frames; receiving a human interface device (HID) signal from the inputting unit; converting the HID signal into an angle signal; and displaying one of the frames of the stereoscopic scene on a displaying unit according to the angle signal; wherein the step of converting the HID signal into the angle signal comprises: obtaining an HID value of the HID signal; normalizing the HID value to obtain a normalized value; and multiplying the normalized value by 90° to obtain the angle signal.
2. The method for displaying the stereoscopic scene according to claim 1, wherein the normalized value is between -1 to 1 .
3. The method for displaying the stereoscopic scene according to claim 1, wherein the step of generating the stereoscopic scene comprises: searching the frames for the stereoscopic scene from a plurality of frame registers; and storing the stereoscopic scene temporarily.
4. The method for displaying the stereoscopic scene according to claim 1, further comprising: when the stereoscopic scene is being generated and an inference procedure needs to be performed, performing the inference procedure by an artificial intelligence acceleration processor.
5. The method for displaying the stereoscopic scene according to claim 4, wherein the artificial intelligence acceleration processor is a neural-network processing unit (NPU) or an intelligence processing unit (IPU).
6. The method for displaying the stereoscopic scene according to claim 4, comprising: loading an inference model to a computing register; directing an address of the computing register to the artificial intelligence acceleration processor; and executing the inference model by the artificial intelligence acceleration processor to perform the inference procedure.
7. An electronic device, comprising: a displaying unit; an inputting unit, comprising a hot key; a stereoscopic scene generating unit, wherein if the hot key is pressed, the stereoscopic scene generating unit generates a stereoscopic scene comprising a plurality of frames; a controlling unit, used for receiving a human interface device (HID) signal from the inputting unit and converting the HID signal into an angle signal; and an imaging unit, used for displaying one of the frames of the stereoscopic scene on the displaying unit according to the angle signal; wherein the controlling unit comprises: an HID value retrieving component, used for obtaining an HID value of the HID signal; a normalization component, used for normalizing the HID value to obtain a normalized value; and an angle conversion component, used for multiplying the normalized value by 90° to obtain the angle signal.
8. The electronic device according to claim 7, wherein the normalized value is between -1 to 1 .
9. The electronic device according to claim 7, wherein the stereoscopic scene generating unit comprises: a searching component, used for searching the frames for the stereoscopic scene from a plurality of frame registers; and a scene register, used for temporarily storing the stereoscopic scenes.
10. The electronic device according to claim 7, further comprising: an artificial intelligence

acceleration processor; and an inference procedure translation unit, wherein when the stereoscopic scene is being generated and an inference procedure needs to be performed, the inference procedure translation unit performs the inference procedure by the artificial intelligence acceleration processor.

11. The electronic device according to claim 10, wherein the artificial intelligence acceleration processor is a neural-network processing unit (NPU) or an intelligence processing unit (IPU).

12. The electronic device according to claim 10, wherein the inference procedure translation unit loads an inference model to a computing register and directs an address of the computing register to the artificial intelligence acceleration processor, so that the artificial intelligence acceleration processor executes the inference model to perform the inference procedure.
